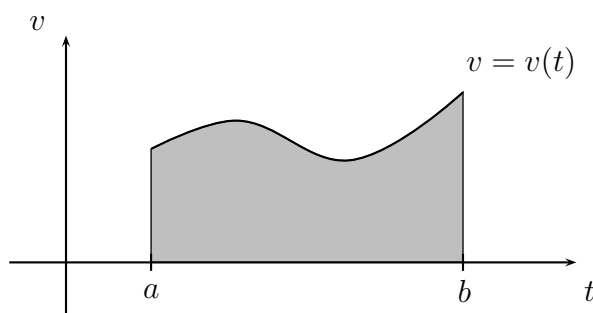


Notes for ‘Inferring displacement from velocity’

Important Ideas and Useful Facts:

- (i) **Area under the velocity curve:** Consider an object or particle moving along a straight line, where one direction is regarded as being positive, and the reverse direction is regarded as negative, and there is some reference point denoted by zero (as in the real line).

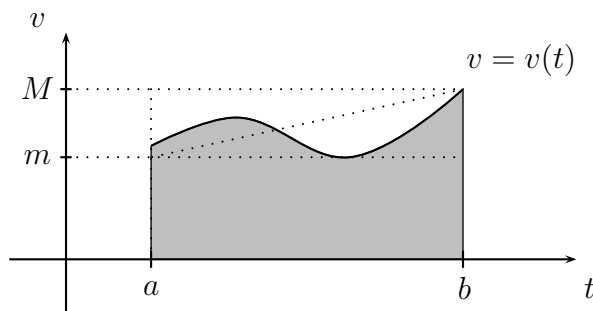
Denote displacement of the object or particle by $x = x(t)$ and its velocity by $v = v(t)$, functions of time t , using compatible units that are appropriate for the particular context.



The area under the velocity curve $v = v(t)$ for $a \leq t \leq b$, where $a < b$, represents the change in position $x(b) - x(a)$.

- (ii) **Upper and lower bounds for the area under the velocity curve over an interval:** As above, suppose m is the minimum and M is the maximum velocity for $a \leq t \leq b$. Then the *lower rectangle* of width $b - a$ and height m is a lower bound for the distance travelled. The *upper rectangle* of width $b - a$ and height M is an upper bound for the distance travelled over this interval. Hence

$$m(b - a) \leq \text{distance travelled} \leq M(b - a).$$



A useful estimate of the distance travelled is the average of the lower and upper bounds:

$$\frac{(m + M)(b - a)}{2},$$

that is, the area of the *trapezium* of base length $b - a$ and vertical side lengths m and M .

Examples:

1. A car travels along a straight road for 50 seconds at a uniform velocity of 95 kilometres per hour. Estimate, to the nearest 100 metres, how far the car traveled.

Solution: There are 1,000 metres in a kilometre and 3,600 seconds in an hour, so the distance travelled is

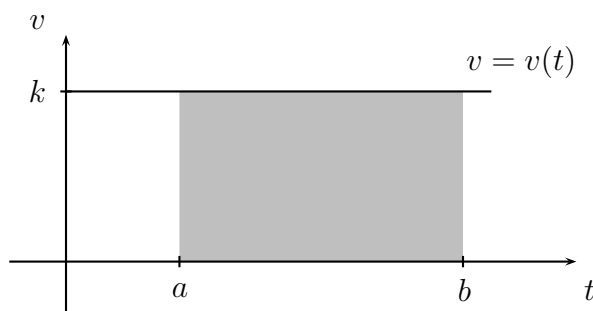
$$\frac{95 \times 1000}{3600} \times 50 = 1,319.4 \text{ m},$$

which we estimate to be 1,300 metres to the nearest 100 metres.

2. If the velocity curve $v = v(t) = k$ is constant then the distance travelled (in appropriate units) for $a \leq t \leq b$ is

$$k(b - a),$$

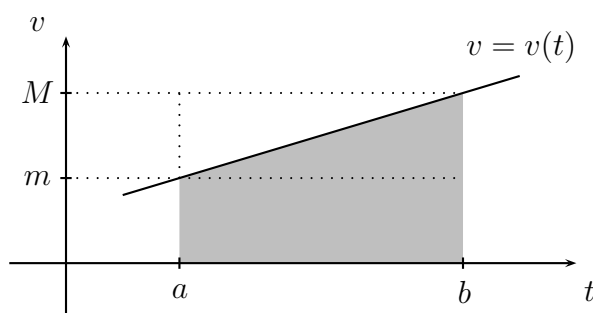
which is the area of the rectangular region under the curve, of width $b - a$ and height k units.



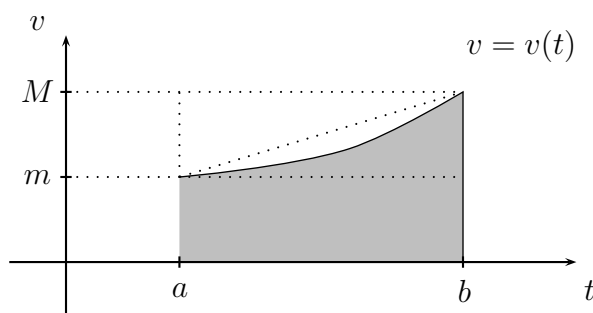
3. If the velocity curve $v = v(t)$ is linear and increasing (constant positive acceleration) then the distance travelled (in appropriate units) for $a \leq t \leq b$ is

$$\frac{(m + M)(b - a)}{2},$$

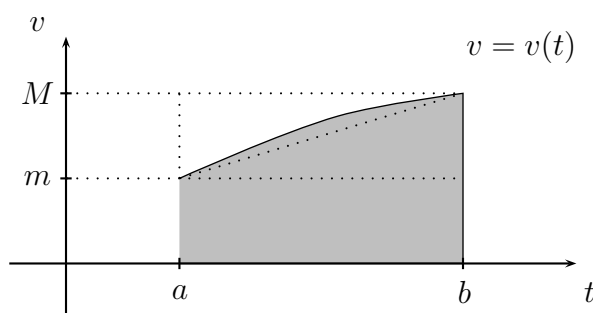
which is the area under the curve, and the average of the lower and upper rectangles of width $b - a$ and heights m and M respectively



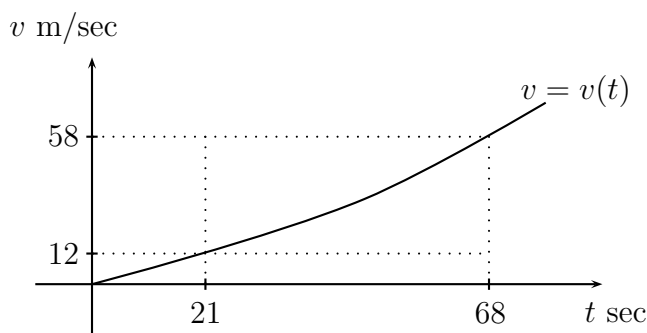
4. If the velocity curve $v = v(t)$ is increasing and concave up then the average of the lower and upper bounds is an overestimate of the distance travelled



5. If the velocity curve $v = v(t)$ is increasing and concave down then the average of the lower and upper bounds is an underestimate of the distance travelled



6. The velocity curve $v = v(t)$ m/sec of an accelerating object moving along a straight line is shown below, as a function of t sec, and is shaped slightly concave upwards.



Find lower and upper bounds for the distance travelled by the object over the time interval $21 \leq t \leq 68$ seconds. Use the average to estimate the distance travelled, and comment about whether this is likely to be an underestimate or overestimate.

Solution: The lower rectangle sitting just beneath the curve has area

$$(68 - 21) \times 12 = 564 ,$$

so a lower bound on the distance travelled is 564 metres. The upper rectangle sitting just above the curve has area

$$(68 - 21) \times 58 = 2726 ,$$

so an upper bound on the distance travelled is 2,726 metres.

The average is

$$\frac{(12 + 58)(68 - 21)}{2} = \frac{564 + 2726}{2} = 1645 ,$$

so a reasonable estimate of the distance travelled should be about 1,645 metres. We expect this to be a slight overestimate, as the curve appears to be slightly concave up.

7. A car accelerates from rest, reaching a speed of 24 m/sec after six seconds. The following velocities are recorded at the end of each second:

time from rest (sec)	0	1	2	3	4	5	6
velocity (m/sec)	0	8	13	17	20	23	24

Use the average of lower and upper rectangular approximation on each time interval, to estimate the distance travelled after the six seconds have elapsed. (You may assume the velocity curve is increasing throughout.)

Solution: Since the velocity of the car is increasing throughout, and the intervals are one second each, a lower bound L for the distance travelled will be obtained by adding up the velocities, in succession from 0 up to 23, since each of these will be a lower bound on the particular interval of one second:

$$L = 0 + 8 + 13 + 17 + 20 + 23 = 81 \text{ m} .$$

An upper bound U for the distance travelled will be obtained by adding up the velocities, in succession from 8 up to 24, since each of these will be an upper bound on the particular interval of one second:

$$U = 8 + 13 + 17 + 20 + 23 + 24 = 105 \text{ m} .$$

The average of these should be a reasonable estimate of the true distance travelled:

$$\frac{L + U}{2} = \frac{81 + 105}{2} = 93 \text{ m} .$$

Thus we expect the car travelled about 93 metres over these six seconds.

8. A car comes to a stop, ten seconds after the driver slams on the brakes. While the brakes are on, the following velocities are recorded:

time since brakes applied (sec)	0	2	4	6	8	10
velocity (m/sec)	34	24	16	9	4	0

Use the average of lower and upper rectangular approximations on each two second time interval to estimate the distance travelled over those ten seconds as the car comes to a halt. (You may assume the velocity curve is decreasing throughout.)

Solution: Since the velocity of the car is decreasing throughout, and the intervals are two seconds each, a lower bound L for the distance travelled will be obtained by adding up the velocities, in succession from 24 down to 0, since each of these will be a lower bound on the particular two second interval, and multiplying through by this common interval length of two seconds:

$$L = (24 + 16 + 9 + 4 + 0) \times 2 = 106 \text{ m} .$$

An upper bound U for the distance travelled will be obtained by adding up the velocities, in succession from 34 down to 4, since each of these will be an upper bound on the particular two second interval, and multiplying through by this common interval length of two seconds:

$$U = (34 + 24 + 16 + 9 + 4) \times 2 = 174 \text{ m} .$$

The average of these should be a reasonable estimate of the true distance travelled:

$$\frac{L + U}{2} = \frac{106 + 174}{2} = 140 \text{ m} .$$

Thus we expect the car travelled about 140 metres over these ten seconds, as it came to a halt.